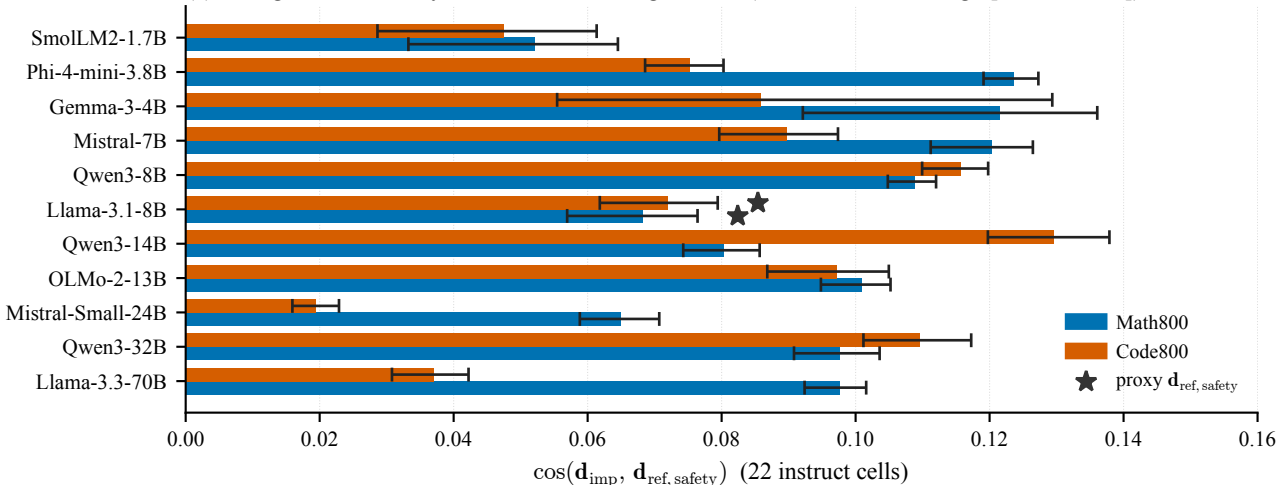


(a) Recognition  $\perp$  safety refusal — 22 main-grid cells (mean  $\cos$  0.087, range [0.020, 0.130])



(b) Pretraining origin — 6 base/instruct pairs ( $\Delta\cos$  range [-0.008, +0.110], mean +0.037)

